

ABDULLAH SHAMAIL

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EDUCATION

Doctor of Philosophy, Computer Engineering

Expected 11/2026

Iowa State University (ISU), Ames, IA

- **Major Professors:** Goce Trajcevski, Ashfaq Khokhar
- **Thesis:** Constraint-Aware Scalable Discovery of Persistent Spatio-Temporal Patterns

Master of Science, Computer Engineering

12/2024

Iowa State University (ISU), Ames, IA

- **Major Professor:** Goce Trajcevski
- **Specialization:** Software Systems

Bachelor of Science, Computer Science

06/2020

Lahore University of Management Sciences (LUMS), Lahore, Pakistan

- **Major Professor:** Muhammad Fareed Zaffar
- **BS Thesis:** Authorship Attribution and Obfuscation

TEACHING & MENTORING EXPERIENCE

Lab Instructor & Graduate Teaching Assistant, ISU

Fall 2026

Introduction to Software Systems for Big Data Analytics & Operating Systems: Principles and Practice

- Designed, delivered, and graded comprehensive laboratory modules to bridge theoretical systems concepts with practical, hands-on implementation.
- Guided a combined cohort of 50+ students, providing individualized technical support, debugging assistance, and academic mentorship.

Guest Lecturer & Graduate Teaching Assistant, ISU

Spring 2025, Spring 2026

Software Evolution and Maintenance

- Mentored undergraduate senior students in tracing, debugging, and understanding large codebases across long-term software lifecycles.
- Evaluated professional-grade written reports and oral presentations on experimental software maintenance.

Lab Instructor & Graduate Teaching Assistant, ISU

Spring 2024, Spring 2025

Software Tools for Large-Scale Data Analysis

- Led weekly laboratory sessions for 30+ students covering external memory processing, large-scale parallelism (Hadoop/Spark), and data formats.
- Bridged theoretical data science and practical applications utilizing a project-based learning framework.

Lab Instructor & Graduate Teaching Assistant, ISU

Fall 2024, Fall 2025

Intro to Computer Engineering and Problem Solving & Digital Logic Circuits

- Delivered foundational instruction to 150+ first-year students on algorithmic problem solving, C/C++, and technical reporting.
- Supervised hardware labs for 25+ students covering Boolean algebra, Finite State Machines, and programmable logic devices.

Graduate Teaching Assistant, ISU
Software Analysis & Verification for Safety & Security

Fall 2023

- Guided 80+ senior and graduate students through threat modeling, vulnerability categorization, and verification algorithms for cyber-physical systems.

Senior Capstone Project Mentor, ISU

Spring 2022 – Spring 2024

- Mentored multiple capstone teams (100+ students) through the Agile Software Development Life Cycle, guiding architectural decisions that culminated in a co-authored student demo paper at ACM SIGSPATIAL 2024.

Guest Lecturer & Lab Instructor, ISU

Spring 2021, Fall 2021

Introduction to Spreadsheets and Databases

- Managed instructional logistics and delivered lab lectures (Microsoft Excel and Access) for 200+ non-technical undergraduate students.
- Developed automated grading scripts to provide rapid feedback and ensure an accessible, inclusive learning environment.

Guest Lecturer & Undergraduate Teaching Assistant, LUMS

Fall 2018 – Spring 2020

Deep Learning, Discrete Mathematics, & Computational Problem Solving

- Managed logistics and lab sessions for foundational to advanced computing courses, instructing cohorts of over 200 students.
- Simplified complex mathematical and architectural AI concepts into accessible modules and implemented "early alert" tutorials for struggling students.

RESEARCH EXPERIENCE

Graduate Research Assistant, ISU

06/2022 – Present

- **SDC Framework & Motif Mining (2026):** Designed the Semantic Diversity in Convoys (SDC) framework (achieving up to 65x computational speedups) and developed large-scale motif-mining pipelines using learned embeddings for scientific event prediction.
- **Bond-Aware Relaxed Moving Clusters (BARMC) (2024–2026):** Integrated spatial proximity metrics with domain-specific semantic rules to enhance clustering techniques, achieving a 1.7x reduction in computation time for complex trajectory data.
- **Automatic Detection of Hydrogen Bonds (2024):** Directed an undergraduate research team through an applied software engineering project to automate the identification of long-lasting interactions in simulations.
- **Molecular Dynamics Data Compression (2022):** Developed a domain-specific compression algorithm that reduced storage requirements by 80% and achieved a 3x speedup in data retrieval for complex simulations.

INDUSTRY EXPERIENCE

Engineering Data Science Intern | Altair Engineering, Troy, MI

05/2025 – 08/2025

- Deployed end-to-end PyTorch and TensorFlow pipelines optimizing GNN and Transformer models, achieving a 10x reduction in Mean Absolute Error (MAE) for spatio-temporal physical simulation data.

Software Developer | Taar Consulting, Lahore, Pakistan

05/2018 – 06/2018

- Designed and integrated a Customer Feedback Plugin into existing transaction systems, capturing feedback from 1,000+ daily transactions to generate actionable business insights.

PUBLICATIONS

Peer-Reviewed Journal Publications

- **Shamail, A.**, Anowar, M., Trajcevski, G., Khokhar, A., Murad, S., Jameson, C., 2026. *Bond-Aware Moving Clusters of Atomic Trajectories with Relaxed Persistency*. *ACM Trans. Spatial Algorithms Syst.* 12, 2, Article 8 (June 2026), 27 pages.
- Anowar, M., **Shamail, A.**, Wang, X., Trajcevski, G., Murad, S., Jameson, C. J., & Khokhar, A. (2024). Compressing generalized trajectories of molecular motion for efficient detection of chemical interactions. *Information Systems*, 125, 102426.
- Mubin, O., Alnajjar, F., **Shamail, A.**, Shahid, S., Simoff, S. *The new norm: Computer Science conferences respond to COVID-19*. *Scientometrics* 126, 1813–1827 (2021).

Conference Proceedings

- **Shamail, A.**, Trajcevski, G., Khokhar, A., Zufle, A. *Semantically Diverse Convoys*. 2026 27th IEEE International Conference on Mobile Data Management (MDM), Athens, Greece.
- **Shamail, A.**, Anowar, M., Trajcevski, G., Khokhar, A., Murad, S., Jameson, C. *Bond-Aware Moving Clusters of Atomic Trajectories with Relaxed Persistency*. *ACM SIGSPATIAL '24*, 605–608.
- Anowar, M., **Shamail, A.**, Albertsen, A., Hall, B., Thielen, T., Riemersma, B., Trajcevski, G. *Detecting and Visualizing Bond-Forming Convoys in Atomic and Molecular Trajectories*. *ACM SIGSPATIAL '24*, 657–660.
- Anowar, M., **Shamail, A.**, Wang, X., Trajcevski, G., Murad, S., Jameson, C. J., & Khokhar, A. (2022). *Generalization Aware Compression of Molecular Trajectories*. *ADBIS 2022*. Lecture Notes in Computer Science, vol 13389. Springer. (Best Paper Award)

HONORS & AWARDS

- **Preparing Future Faculty Associate:** Awarded by Iowa State University (2025).
- **NSF Student Travel Grant:** Awarded through a competitive selection process (2024).
- **Best Paper Award:** *Generalization Aware Compression of Molecular Trajectories* at ADBIS (2022).

SERVICE & LEADERSHIP

- **Volunteer, ACM SIGSPATIAL (2024):** Coordinated logistics for 300+ attendees and facilitated technical and poster sessions in Atlanta, Georgia.
- **President, Pakistan Student Association, ISU (05/2023 – 05/2024):** Managed budgets and led a data-driven social media campaign raising \$20,000 for flood relief efforts.
- **Vice President, Index: Design and Innovation Society, LUMS (05/2019 – 05/2020):** Mentored 50 students and successfully organized UX Pakistan, a global design conference for 500 attendees.

SOFTWARE & CODING SKILLS

- **Data Analysis & ML:** TensorFlow, PyTorch, Sklearn, Keras, Pandas, Numpy, Apache Spark, Apache Flink, Hadoop, Pig Latin, SQL, MATLAB, SQL Server, MDX, MySQL.
- **Core Languages:** Python, C++, Objective C, JavaScript/ReactJS.
- **Miscellaneous:** LaTeX, GitHub, Trajectory Analysis, MS Excel, MS Access.

REFERENCES

- **Ashfaq Khokhar**, Professor, Iowa State University | ashfaq@iastate.edu | 515-294-1097
- **Goce Trajcevski**, Associate Professor, Iowa State University | gocet25@iastate.edu | 515-294-2176
- **Mani Mina**, Teaching Professor, Iowa State University | mmina@iastate.edu | 515-294-3918
- **Karen Menzel**, Program Specialist III, Iowa State University | melig@iastate.edu | 515-294-4531